

Some Innate Characteristics of Neural Network Models of Morphological Inflection

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Abstract

Neural Network Models of Morphological Inflection (NNMIs) have seen impressive improvements over the last several decades. These improvements, however, lie in raw accuracy and other practical issues. When the performance of these models is instead compared to that of child learners, many of the shortcomings of the classic models persist in today’s NNMIs: **(1)** they struggle with generalization across sparse paradigms **(2)** they are prone to over-irregularization, and **(3)** they do not follow child acquisition trajectories. The persistence of these issues suggests that they reflect inherent or “innate” characteristics of NNMIs as a class which are distinct from the characteristics of human language learners. As such, even substantial practical improvements do not necessarily entail increased cognitive plausibility.

1 Introduction

For nearly forty years now, Neural Network Models of Morphological Inflection (NNMIs) have played a major role in discussions surrounding not just the nature of natural language morphology, but broader linguistic theory, and even the fundamental nature of cognitive representations. Since Rumelhart and McClelland (1986, henceforth RM86) and Pinker and Prince (1988) kicked off the Past Tense Debate of the 1980s and 1990s (see McClelland and Patterson 2002, Pinker and Ullman 2002, Pinker 2006, and Seidenberg and Plaut 2014 for surveys), NNMIs have continued to improve. NNMI architectures have developed from simple feed-forward networks through to RNNs and transformers. These have successively shown improvements, both in their raw performance and in their generality and practical applicability. This naturally raises the question: do the practical improvements of modern NNMI architectures strengthen the position of NNMIs as cognitive models? In a paper explicitly aimed at reopening the Past Tense Debate, Kirov and Cotterell (2018, henceforth KC18) argued that they do; a number of replies ensued (Corkery et al., 2019; McCurdy et al., 2020a,b; Belth et al., 2021, i.a.).

Around the same time, there was substantial interest in defining standardized shared tasks for morphological inflection (Cotterell et al., 2016, 2017, 2018; McCarthy et al., 2019; Vylomova et al., 2020; Pimentel et al., 2021; Kodner et al., 2022; Goldman et al., 2023). Though these shared tasks have focused on overall accuracy over typologically diverse languages and not primarily on modeling language acquisition, they have spurred on the development of new NNMIs, as well as a standardization of the task design for these models: modern NNMIs solve a string-to-string mapping problem from lemmas to inflected forms, typically with the addition of some features indicating something about the syntax-semantics of the word to be inflected. For example, a model may receive as input the string `walk, V;PST;3;SG` and be expected to produce as output the string `walked`; NNMIs vary in how they encode the features as a part of their input. NNMIs in the early days of the Past Tense Debate also solved inflection problems, but conceived of the task in a wider variety of ways, including lemma-to-inflectional pattern or as a multiple choice problem due to technical limitations of the time.

The NNMIs of the Past Tense Debate and their modern descendants operate on Marr’s *computational level* of explanation in cognitive science (Marr, 1982, ch. 1). The computational level requires a clear specification

Long-Tailed Distributions in Child-Directed Speech

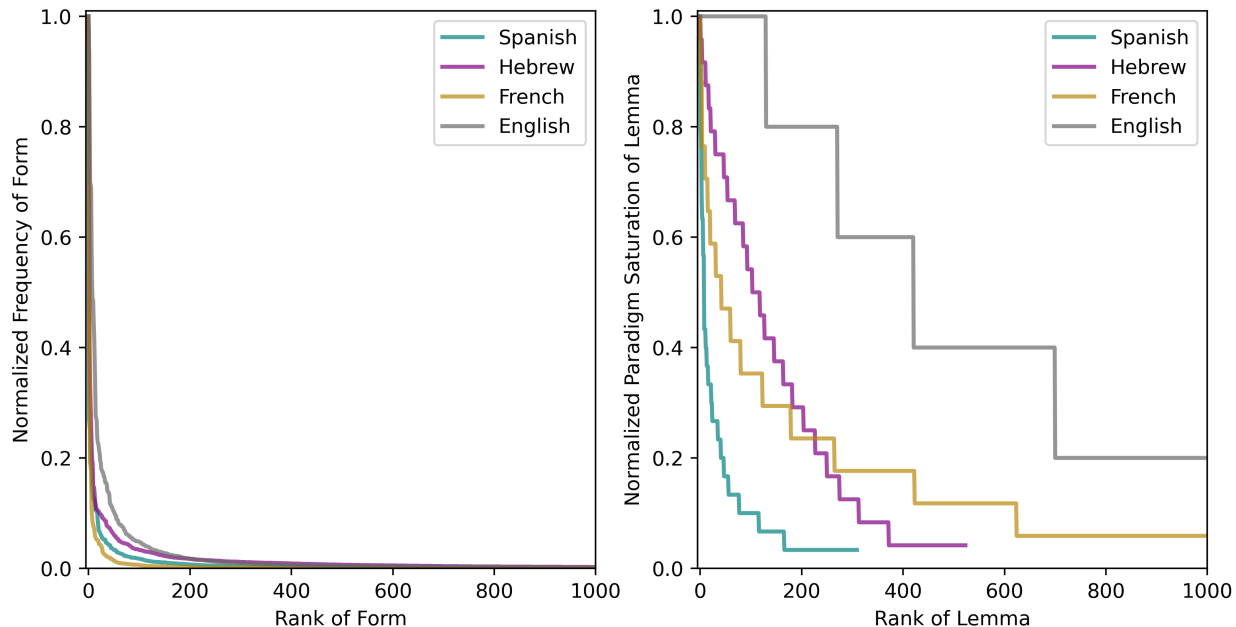


Figure 1: The long-tailed distributions of frequency (left) and paradigm saturation (right) for CHILDES. Both plots are normalized by dividing by the maximum attested value for each language in CHILDES.

of the input, a clear specification of the output — which includes things like accuracy, error rates and types, and developmental trajectories — and the logic or strategy behind the mapping from the input to output. Indeed, to more seriously investigate the suitability of NNMI for language acquisition, we must look beyond raw accuracy: a good computational model of human language learning and representation should in some way behave like a human, a point on which researchers on both sides of the original debate agreed.

Our goal is not to provide a comprehensive look at all aspects of the Past Tense Debate: distributed vs. symbolic computations, single- or dual-route models of processing, the biological plausibility of neural networks, and so on. Rather, we target our comparison at three non-human-like behaviors (outputs in the context of an input) that were investigated in the NNMI of the original debate and persist in deep learning-era descendants: these models struggle to generalize across sparse paradigms (§2), are prone to developmentally strange over-irregularization (§3), and have yet to reproduce developmental regressions (§4). While in decades past, these shortcomings could reasonably be dismissed as temporary deficits of early NNMI architectures, their persistence into the modern era, where NNMI otherwise achieve very high accuracy, suggests another possibility: they constitute “innate” properties of NNMI as a class of morphology learner, which have persisted across a vast range of increasingly complex neural architectures. The takeaway is clear: even substantial advancements in neural architectures and their concomitant performance improvement does not necessarily entail increased cognitive plausibility.

2 Generalization from Sparse Input

It is well known that the frequency of words in linguistic data—including child-directed speech—follows a highly skewed Zipfian distribution, with a word’s frequency being inversely proportional to its rank (Zipf, 1936). Since frequency is correlated with order of acquisition (Palermo and Eberhart, 1968; Goodman et al., 2008; Swingley and Humphrey, 2018; Braginsky et al., 2019), learning requires generalization from frequent forms, which are learned earlier, to less frequent ones.

Beyond vocabulary, children also face sparse input in terms of morphological paradigms. This can be

measured in terms of PARADIGM SATURATION, which is the proportion of a given lemma’s possible inflected forms that actually occur in some corpus, such as the child’s input (Chan, 2008). When sorted by their paradigm saturations, lemmas follow highly skewed distributions that scale up with the size of the paradigm. This derives, at least in part, from the proliferation of Zipfian and other long-tailed frequency distributions in corpora. If a lemma only appears a handful of times, and its potential paradigm is large, it can only occur in a handful of its possible forms. For example, measuring paradigm saturations in the Spanish CHILDES corpora (MacWhinney, 1996), Lignos and Yang (2016) found that the most saturated verb reaches 83%, while the average verb appears in just 7.9%, or approximately 4 out of the 55 possible inflected forms. Figure 1 visualizes the long-tailed distributions of frequency (left) and paradigm saturation (right) in samples of child-directed speech (CDS) extracted from CHILDES (MacWhinney, 1996) and intersected with UniMorph 3 (McCarthy et al., 2020) to normalize the tags and filter out errorful annotations.¹

Because of the sparse and skewed distributions of lemmas and inflected forms in child-directed speech, morphological acquisition is at its core a matter of *generalization*. Since most words, and therefore inflected forms, that are attested at all in the input are only attested once or twice, and even robustly attested lemmas are only attested in a fraction of their possible forms when paradigms are large, children must both generalize across lemmas and often across inflected forms to fill in parts of paradigms that will not be evidenced directly in their input.²

It is typical in computational modeling of morphological acquisition to model vocabulary size in terms of types “known” to the learner rather than the number of tokens in the input. This corresponds to the observation, from a wide range of linguistic perspectives, that it is *type* frequency in a speaker’s lexicon that drives generalization, and that morphological acquisition occurs over inflected forms of *known lemmas* in the learner’s lexicon (e.g., Aronoff, 1976; MacWhinney, 1978; Bybee, 1985; Plunkett and Marchman, 1991; Baayen, 1993; Elman, 1998; Pierrehumbert, 2002; Yang, 2016). As such, the input to the model during training is an attempt to directly inject an appropriately sized lexicon (but see the discussion of developmental regression in RM86 in §4).

Cross-linguistically, children learn and accurately produce most of their native language’s morphology before school age. Since the input to morphological acquisition is inflected forms of lemmas in the child’s lexicon, receptive vocabulary gives the best measure of lexicon size. However, measuring receptive vocabulary is quite challenging: while many authors report productive vocabularies of 500-1,000 words around age three (Fenson et al., 1994; Hart and Risley, 1995; Bornstein et al., 2004; Szagun et al., 2006), there are no figures on receptive vocabulary at precisely this age. Fenson et al. report receptive vocabulary to be between 150-200 words at 16 months, while Segbers and Schroeder (2017) report about 6,000 words in first grade; Shipley and McAfee (2023)’s SLP evaluation resources expect three- to four-year-olds to have a receptive vocabulary of “1,200 to 2,000 or more” and productive vocabulary of “800 to 1,500 or more” words. While there is a great deal of variation in these numbers, it is important to note that they all represent some measure of the child’s *entire* vocabulary, while morphological paradigms target individual parts of speech. Bornstein et al. (2004) measure the distribution of parts of speech in early child vocabulary, and find that while nouns and verbs predominate, the ratio varies over time, with neither even reaching 50%. Taken together, this evidence suggests that nominal and verbal morphology is acquired on the basis of only several hundred known word types.

Since children must accurately perform generalizations to learn from their small and skewed input, we should expect the same from a computational model of language acquisition. Indeed, the surprising output accuracy of early connectionist models such as RM86 trained on a few hundred word types – computing power limited them to small data by necessity – was taken as evidence of the plausibility of these early NNMs. More recently, KC18 claim that their encoder-decoder RNN can learn the English past tense from developmentally plausible input. However, they train their model on over 3,500 verb lemmas, an order of magnitude larger than the number of lemmas from which English-learning children acquire the past tense or the 420 verb lemmas on which RM86 was trained (§4).

¹The denominator in the Chan (2008) and Lignos and Yang (2016) is the a paradigm’s size according to a language’s grammatical description. Payne (2022) presents an alternative denominator, which is the product of the number of dimensions of all features. This counts all possible feature combinations including those that are not specially marked in the language. While the former approach, which we adopt here, is a useful corpus description, the latter approach is meant to represent a learner’s view of the language before they have learned which feature combinations are marked.

²Some researches have argued that rather than merely posing a challenge to learning, sparse skewed input distributions actually facilitate learning (Chan, 2008; Janda and Tyers, 2021).

Further, most early NNMI were only designed to learn just one morphological pattern, such as the English past tense, rather than an entire paradigm (but see MacWhinney and Leinbach 1991 and Plunkett and Juola 1999). However, the popularization of classes of string-to-string mapping NNMI in the modern era and inclusion of features as a part of model input has seen the deployment of models in-principle capable of learning all parts of a paradigm simultaneously. KC18 followed up on their spiritual replication of RM86’s English past-only experiments by training their model on full English verbal paradigms, again achieving good performance. They argue that this is a strong point in favor of their model’s cognitive plausibility. Yet, of the 3,144 most frequent verb stems in English CHILDES, only 656, less than a quarter, appear in their complete paradigms. By extension, one would have to learn over 14,000 verbs in order to hear 3,500 in their complete paradigms, two orders of magnitude greater than the number of verbs an English-learning 3-year-old knows. Though practical improvements in computing power allow for modern NNMI can be trained on orders of magnitude more data than RM86, this has led to an epidemic of loose definitions for developmentally “plausible” inputs, both in terms of size and saturation: from KC18’s 3,500 complete paradigms and Dankers et al. (2021)’s 46,000 noun plurals to Warstadt and Bowman (2022)’s 100 million word tokens for syntactic language modeling.

Given the long-tailed nature of paradigm saturation, a developmentally-plausible approach to learning multiple pieces of the paradigm must do so from incomplete paradigms. This is a particular issue for languages with larger paradigms than English, since as paradigm size increases, so too does sparsity and the need to rely more heavily on generalization across pieces of the paradigm. This is illustrated on the right side of Figure 1: paradigm saturation drops off much more quickly for Spanish than for French, which has a smaller paradigm than Spanish, or English, which has a smaller paradigm still (cf. Legate and Yang, 2007). We refer to this type of generalization as “feature-based generalization,” since pieces of the paradigm are generally represented with sets of semantic features (e.g. {PRS, 3, SG}) in the modern NNMI literature. In highly-inflected languages, the sparsity of the paradigms is extreme and may require generalizing to slots in which *no* lemma has been seen. For the purposes of illustration, imagine a child learning Spanish who has never seen the imperfect past second singular form of *any* lemma, but can use feature-based generalization to form it: if they have seen *ama-rá-s* (*love*-FUTURE-2.SG) and *ama-ba-n* (*love*-IMPERFECT-3.PL), for example, they may learn that *-ba-* realizes the imperfect past, *-s* realizes the second singular, and thus that the second singular imperfect past of *ama-r* is *ama-ba-s*. The nature of the input to children learning highly inflected languages entails that they must make such generalizations in order to acquire the complete morphological system.

Yet, NNMI largely fail to make feature-based generalizations, even on highly agglutinating languages where this is both developmentally necessary and in-principle feasible, since inflected forms are predictable from their component features. The models investigated in Kodner et al. (2023c) – a character transformer from Wu et al. (2021) and two variants of a character transducer from Wehrli et al. (2022) – showed a massive performance difference between test words with feature sets that were seen in training and those with novel feature sets. Even when trained on 1,600 items, for example, the transformer suffered a 39.9-point performance drop on test items with novel feature sets, while the best variant of the transducer saw a 52.1-point drop. Kodner et al. (2023a) designed linguistically-motivated probes to test specific instances of in-principle feasible feature-based generalization. Testing the same two models as well as KC18 on English (not agglutinating), Spanish and Swahili (highly agglutinating), they find that the transformer and transducer were able to achieve feature-based generalization on the simplest probes but struggled on more complex but in-principle feasible probes across all three languages. In contrast, KC18 showed no ability at all for feature-based generalization.

The key task for a child acquiring morphology is that of generalization from small, sparse, and skewed input. If an NNMI is to be taken as a plausible model of morphological acquisition, then, it must also generalize and achieve similarly high accuracy from similarly sparse input. While high performance is generally reported for modern NNMI, they are typically trained on unfairly large or saturated data. Trained on more plausible input in terms of distribution and training size, their performance and generalization abilities both drop (Belth et al., 2021; Kodner et al., 2023c). But raw accuracy isn’t everything: even if they achieved human-like accuracy, the consistent failure of NNMI to replicate human-like errors continues to hamper potential claims of their cognitive plausibility.

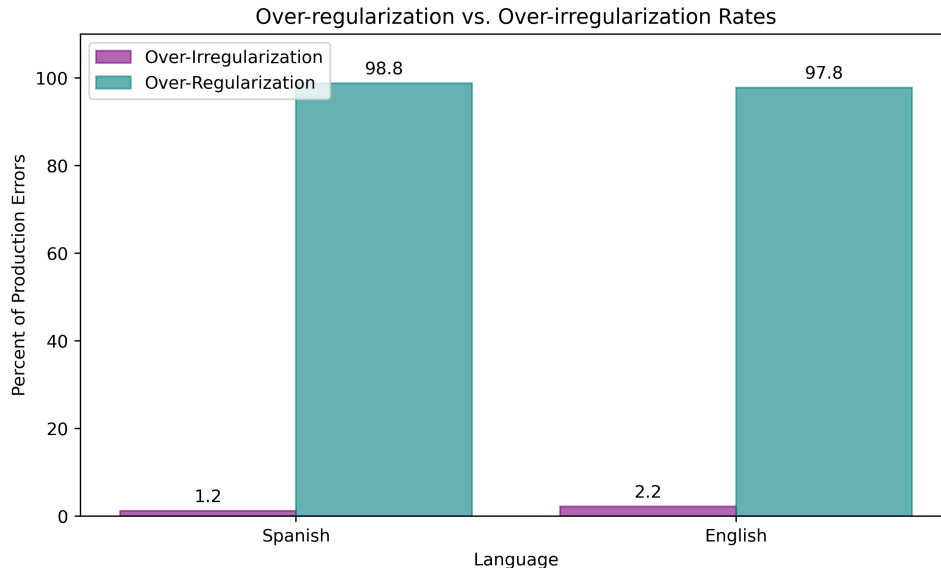


Figure 2: A visualization of the asymmetry between over-regularization and over-irregularization errors in children acquiring English (Xu and Pinker, 1995; Maslen et al., 2004) and Spanish (Clahsen et al., 2002) verbs.

3 Over-Regularization vs. Over-Irregularization

Children’s production errors show a strong asymmetry between *over-regularizations* (extensions of productive patterns, e.g., **go-goed*) and *over-irregularizations* (extensions of irregular patterns, often from phonologically similar words e.g., **wipe-wope*). Over-regularizations make up 97.8% of English past tense production errors and 98.8% of Spanish finiteness marking production errors (Figure 2; Xu and Pinker 1995; Maslen et al. 2004; Clahsen et al. 2002). Several further studies have observed similar patterns for English (Palermo and Eberhart, 1968; Marcus et al., 1992; Maratsos, 2000; Yang, 2002), Spanish (Mayol, 2007), German (Clahsen et al., 1992; Clahsen and Rothweiler, 1993; Elsen, 2002; Szagun et al., 2006; Kauschke et al., 2011), Italian (Caprin and Guasti, 2009), Swahili (Deen, 2005) and other Bantu languages (Demuth, 2003), Hebrew (Lustigman, 2013; Bat-El, 2014), Arabic (Ravid and Farah, 1999), and Inuktitut (Allen, 1996). The asymmetry between over-regularization and over-irregularization is also evidenced by *wug* tests (following Berko 1958), in which children are asked to inflect nonce forms. While children readily extend productive, regular processes to these nonce forms, they do not extend irregular processes in the same way. Strikingly, only one child of 86 in the original Berko (1958) study analogized *bing* to *bang* or *gling* to *glang* along the lines of phonologically similar verbs like *sing-sang*, while the rest provided *-ed* forms or refused to answer.

NNMIs have failed to capture this asymmetry since the beginning. After training their model on 336 regular verbs, RM86 evaluated it on 72 regular forms. Pinker and Prince (1988) find that of these 72 forms, 18 (25%) yielded unnatural over-irregularizations or double markings/blends (e.g., **mail-membled*, **sip-sept*, **hug-hug*, **type-typeded*). Similar criticisms emerged throughout the Past Tense Debate (Prasada and Pinker, 1993; Xu and Pinker, 1995; Marcus, 2003). Marcus (2003) found that the NNMIs of Plunkett and Marchman (1993) and Daugherty and Hare (1993) both produced more blends combining *-ed* with an irregular stem change than they did over-regularizations for novel verbs, while the models of Hare et al. (1995) analogized to similar stem changing patterns 39% of the time, to dissimilar stem changing patterns 10% of the time, and regularized 51% of the time in their model of Old English.

Despite improvements in raw accuracy, modern NNMIs still fail to capture this asymmetry. While KC18 demonstrate that their encoder-decoder makes only over-regularization errors on irregular test items, it also over-irregularizes regular test items, often with spurious vowel changes not dissimilar to those of RM86 (e.g., **thin-thun*). Indeed, Corkery et al. (2019) found that KC18 overproduces irregulars when compared

to human nonce judgments, leading to a low correlation between the model and human judgments: while humans preferred an irregular for only one nonce word (*chind-chind*), KC18 prefers an irregular for 4, or 7% of the nonce words. Similarly, both of the models evaluated by Gorman et al. (2019) – the RNN of Bergmanis et al. (2017) and the encoder-decoder of Makarov et al. (2017) – make a number of what Gorman et al. term “silly” errors on 11 of the 12 languages tested. “Silly” errors are those which “defy any purely linguistic characterization,” such as RM86’s **mail-membled* (Gorman et al., 2019, p. 142).

Finally, Kodner et al. (2023b) find that the Wu et al. (2021) character transformer and Wehrli et al. (2022) transducer both fail to replicate the asymmetry between over-regularization and over-irregularization for the English past tense and Arabic noun plurals. In English, while over-regularizations at least made up a slight majority of production errors for the Wehrli et al. model, they were in the minority for the Wu et al. model, which produced a plethora of over-irregularizations and nonsense predictions. Across models, most Arabic errors were over-applications of unproductive broken plural patterns to other broken plurals (e.g., **nabiyy-nab* instead of *?anbiyā?*) or to regular, sound plurals (e.g., **mafrūb-mafārīb* instead of *mafrūbāt*). This is in contrast to the over-application of the productive, feminine sound plural to both broken and masculine sound plurals by children (Ravid and Farah, 1999), but is comparable to the preference for over-application of the broken plural exhibited by earlier NNMI (Plunkett and Nakisa, 1997).

A critical weakness of the English past tense as a test case is the fact that the default *-ed* pattern is also overwhelmingly the most frequent. Thus, an NNMI’s preference for *-ed* over over-irregularizations could be argued to be driven by frequency matching. Indeed, in Hare et al.’s (1995) study based on Old English, where “weak” *-ed* verbs make up a smaller proportion of past forms compared to Modern English, “strong” stem-changing patterns have greater pull on model predictions. German noun plurals take this much farther with their minority default *-s* (Mugdan, 1977; Schöler and Kany, 1989; Clahsen et al., 1992, 1996), occurring with less than 5% of nouns (Clahsen et al., 1992; Marcus et al., 1995; Wiese, 1996; Sonnenstuhl and Huth, 2002). In their analysis of the German noun plural system, Marcus et al. (1995) make the strong claim that RM86 – and indeed, any connectionist model – will fail to extend the productive *-s* and will instead overextend the frequent *-e* or *-(e)n*. Subsequent work did not test this idea directly on the NNMI of the time. While Hahn and Nakisa (2000) train two simple 3-layer NNMI with backpropagation on German pluralization data, reporting over 80% accuracy for both, they did not provide error analysis that would allow us to test Marcus et al.’s claim.

Nevertheless, Marcus et al.’s pronouncement holds true a quarter century later. McCurdy et al. (2020a) compare human-predicted inflections and KC18-predicted inflections on 24 nonce plurals taken from Marcus et al. (1995), all presented with neuter gender. They find that KC18 indeed has an overwhelming preference for the frequent *-e* suffix: 80% of its predictions take *-e* vs. just 45% of human predictions. The model also struggles to correctly apply *-s*, with an F-score of 0.6 for this suffix, even though its F-score on the comparably rare *-er* is 0.84; there is no significant correlation between KC18 and human judgments for *any* of the German suffixes. Building on this, McCurdy et al. (2020b) find that KC18 relies much more heavily on grammatical gender to predict the plural affix than humans, who seem to rely more on phonological cues. Kodner et al. (2023b) also test the Wu et al. (2021) and Wehrli et al. (2022) models on German noun plurals, finding that both over-apply the *-s* affix to some degree, and Wehrli et al. (2022) overextends it for the developmentally-consistent training sizes of 300 to 400 items. However, over-application of the frequent *-e* is by far the dominant error for all models when trained on the full 600 training items, and the error rate remains unacceptably high. Even the RNN of Dankers et al. (2021), designed specifically for applications to German noun plurals, is biased towards *-e* and does not handle low-frequency plural classes well.

Though the asymmetry between over-regularization and over-irregularization in acquisition is widely attested cross-linguistically, no NNMI has succeeded in replicating it. The consistent, implausibly high rates of over-irregularization by NNMI suggests that their mechanism of generalization is fundamentally different from that of children, and their performance on German noun plurals provides a clue as to where this difference lies: NNMI are innately sensitive to *frequency* rather than *productivity*.

4 Acquisition Trajectories

Over-regularization errors are also implicated in *developmental regression* (or “*U-shaped*” learning), outwardly non-monotonic learning trajectories in language acquisition caused by a transient drop in production accuracy

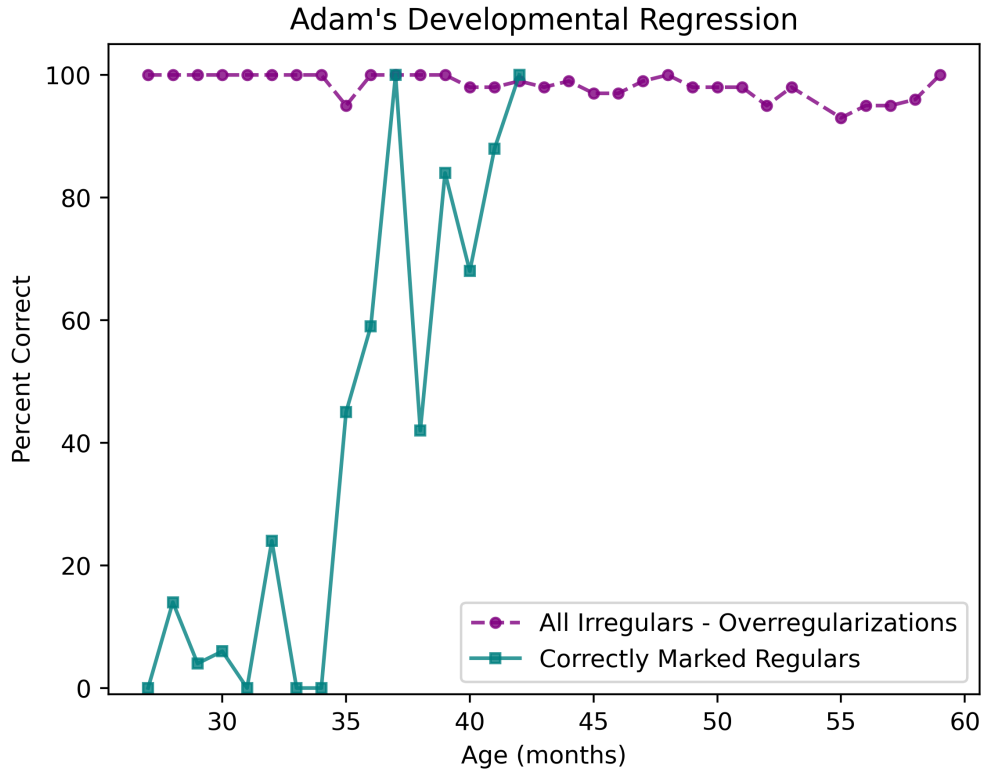


Figure 3: Proportion of regular verbs that were correctly marked as opposed to unmarked (teal) and proportion of irregulars subject to overregularization subtracted from 100% (purple) for the child “Adam” studied by Brown (1973). This is a reproduction of plot reported by Marcus et al. (1992).

(Ervin and Miller, 1963; Bowerman, 1982). In morphology acquisition, it has been observed for individuals acquiring the English past tense (Marcus et al., 1992), Arabic noun pluralization (Ravid and Farah, 1999), and Spanish verbal inflection (Clahsen et al., 2002), among others. It is typically caused by over-regularization errors decreasing accuracy among irregular productions (Marcus et al., 1992; Clahsen et al., 2002, i.a.).

Under traditional accounts, developmental regression is explained by three developmental phases: during the first phase, the child simply produces memorized inflected forms, leading to high accuracy for these forms, but also a great deal of omission errors due to the sparsity of the input (§2). Figure 3 provides a visualization of developmental regression by the child “Adam” studied by Brown (1973) as later analyzed in Marcus et al. (1992) as he acquired the English past tense. We see that for Adam, this stage lasted until around his third birthday, and is marked by low production accuracy on regulars (teal) but no over-regularization errors for irregulars (purple). During the second phase, the child acquires the relevant productive rule, leading to a jump in regular production accuracy and an increase in over-regularization errors when this productive process is over-applied to irregulars for which the correct inflected form is not yet known. This can be seen clearly at about 35 months in Figure 3: Adam’s regular production accuracy rapidly increases at this time, and he produces his first over-regularization, **feeled*, causing a drop – and thus “regression” – in his irregular production accuracy (Marcus et al., 1992). Finally, as the child learns the correct inflected forms for the irregulars, irregular production accuracy increases, and the child escapes the developmental regression. We can see that Adam’s over-regularization rates seem to be tapering from his regression at around 55-60 months of age.³

The failure of NNMI to reproduce a developmental regression on the English past tense has been a

³While the onset of the developmental regression is clear in this particular plot, its resolution is not. It must have occurred for Adam beyond 60 months.

sticking point since RM86 claimed to have achieved it in their original paper. They did indeed achieve a *U*-shape (much more dramatically than Adam’s), but Pinker and Prince (1988) point out that this was only achievable by “forcing” it through modifying input distributions. RM86 first exposed their model to a set of 10 verbs for 10 epochs, 20% of which were regular, which prevented the acquisition of *-ed*, corresponding to the first stage. They then flooded it with 420 verbs for 190 epochs, 80% of which were regular, which forced over-regularization, corresponding to the second stage. RM86 differs from most later models in that they meaningfully interpret training epochs as “learning episodes.” They justify their approach as modeling growth in the learner’s vocabulary over time; the dramatic change in the input distribution is meant to represent a period of massive vocabulary growth, since “the child learns first about the present and past tenses of the highest frequency verbs; later on, learning occurs for a much larger ensemble of verbs, including a much larger proportion of regular forms” (Rumelhart and McClelland, 1986, 241). However, as Pinker and Prince, and later Marcus et al. (1992, ch. 5) discuss, children who experience a period of massive vocabulary growth do so closer to age 2, well before the onset of *U*-shaped learning. Furthermore, the fact that irregulars are clustered among the high frequency verbs in English (Ullman, 1999; Bybee, 2006) does not imply a change in input distribution over time, and certainly not the drastic one that Rumelhart and McClelland suggest (Pinker and Prince, 1988). Pinker and Prince note that RM86’s observed *U*-shape is particularly dramatic, with a cliff-like accuracy drop exactly when the system is first exposed to the regular-heavy second data set; the developmental regression is thus driven by the sudden change to the model’s input rather than emerging as an “endogenous” property of the model.

Decades later, KC18 also claimed that their model achieves *U*-shaped learning on the English past tense. Following the observation that word types rather than tokens matter for morphological generalization, they do not attempt to train on high frequency words for more epochs. They do, however, interpret performance across training epochs in terms of learner development. On closer inspection, what KC18 describe as *U*-shaped is actually small oscillations in performance over these training epochs. This interpretation has two problems. First, what is typically described as *U*-shaped learning unfolds over months or years; it is not a description of small noisy oscillations. This unusual description of “micro *U*-shaped learning” comes from Plunkett and Marchman (1991). Second, KC18 interpreted their input in terms of types, rather than tokens, so it is unclear why they treat the passes over the input (additional tokens) as a proxy for development. When KC18 is instead trained on incrementally larger input sets as a more plausible proxy for vocabulary growth, there is no evidence of the expected developmental regression for the English past tense (Belth et al., 2021).

Kodner et al. (2023b) also test the Wehrli et al. (2022) model for developmental regression by training on incrementally larger sets. While the model does exhibit non-monotonic learning, this is due to a jump in *over-irregularization* errors at 300 training items, the opposite of the cause of developmental regression in English-learning children. They also investigate whether any of the NNMI’s studied can exhibit the two types of developmental regression found by Ravid and Farah (1999) in Palestinian Arabic-learning children. These regressions result from the two types of over-regularization errors discussed in §3: the first from the over-application of the feminine plural to masculine sound plurals, and the second from the over-application of feminine plural to broken plurals. Kodner et al. find no evidence of either type of developmental regression: for all NNMI’s studied, error rates decrease near-monotonically.

Individual child learners exhibit a range of acquisition trajectories, conditioned both by the phenomenon being learned and the specifics of the individual input. While many English-learning children will exhibit *U*-shaped learning in the past tense, for example, not all do (Marcus et al., 1992; Braginsky, 2022). As such, we should not be surprised if any given NNMI does not exhibit developmental regression on some particular data set. Yet, the consistent failure of *any* class of NNMI to exhibit developmental regression on *any* developmentally realistic data set for *any* language where it is expected over the last nearly 40 years suggests a deeper problem: NNMI’s do not learn in the same way as children.

5 Discussion

The Past Tense Debate of the 1980s and 1990s spanned a slew of contentions. The most important of these, according to the leading contributors of both sides of the debate, was the conflict between symbolic or rule-based models of cognitive representations and connectionist or parallel distributed processing models of cognitive representations; NNMI’s are implementations of the latter. While this is the focus of our article, it

was not the only contention of the debate with lasting impact. Much has been written about the conflict between single-route models of morphological inflection (including most NNMI’s then and now, but see Westermann and Goebel 1995, Westermann 1999, and Hahn and Nakisa 2000) and dual-route models (e.g., Pinker, 1991, 1999; Albright and Hayes, 2003; Yang, 2016), and the neurobiological basis of these models in retrospectives on the Past Tense Debate (McClelland and Patterson, 2002; Pinker and Ullman, 2002; Pinker, 2006; Seidenberg and Plaut, 2014).

Since early NNMI’s were by necessity small and simple neural networks, it was always possible that deficits in their performance could be attributed to their specific implementations. Nearly forty years on, in an era of deep learning, LLMs, and generative AI, modern NNMI’s are far more elaborate, far easier to use, and far more powerful than those of decades past. They generally achieve high output accuracy on a variety of morphological inflection tasks, as have their cousins on other linguistics tasks.

It is all the more striking, then, that modern deep NNMI’s, across architectures, do not consistently outperform classic NNMI’s in all ways. They can fill in paradigms now but struggle when they are fed paradigms following the distributions attested in children’s input. They are still overly focused on superficial properties of the lexicon *per se*, such as frequency and phonological similarity, rather than any underlying notion of productivity, so they still do not replicate the stark asymmetry between over-regularization and over-irregularization in child production errors. Their learning curves still lack developmental regressions even under circumstances where many children exhibit them.

Granted, there are many complexities to model training and the basic task setup which may cloud the true capabilities of NNMI’s. Most obviously, children do not learn morphological inflection in a supervised fashion from a sequence of input-output pairs. While we advocate for advancing this learning task towards a more developmentally realistic setting, our survey here has focused on the majority of studies which perform the standard inflection task. Alternatives which evaluate different aspects of morphological learning, such as morpheme segmentation (Kann et al., 2018; Liu and Prud’hommeaux, 2022), lemmatization in context (McCarthy et al., 2019), and, unsupervised paradigm completion (Kann et al., 2020; Wiemerslage et al., 2021) have been undertaken by NNMI’s in recent years as well. Additional work has studied the effect of tokenization on these models (Nzeyimana and Niyongabo Rubungo, 2022; Nair and Resnik, 2023; Fusco et al., 2024). Furthermore, while we have focused on the similarity in outputs between forty years of models with vastly distinct neural architectures, it is not the case that all models perform equally well or poorly in all cases. A future survey should consider the impact of choice of task and architecture.

Nobody can provide a negative existence proof outside of logic and mathematics, but the lack of progress on NNMI’s’ shortcomings as models of morphological acquisition *despite* massive advancements in so many other aspects of neural network design, training, and deployment, strongly suggests that these problems are not an artifact of a specific implementation. Rather, they are inherent or “innate” to neural network models of morphological inflection. Put simply, NNMI’s constitute a different class of language learner than the human child.

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